

INTERQUANTILE RANGE

Purpose: To identify the presence of outliers in a dataset.

Formula:

- $IQR = Q_3 - Q_1$
- Lesser Outlier = $Q_1 - 1.5 * IQR$
- Greater Outlier = $Q_3 + 1.5 * IQR$

Why 1.5 is used in IQR?

Because it almost near to $\pm 3\sigma$ used in normal distribution and it works well even when data is not perfectly normal.

	Minimum	Q_1	Median	Q_3	Maximum
Day	32	56	74.5	82.5	99
Night	25.5	78	81	89	98

DAY:

$$IQR = 82.5 - 56 = 26.5$$

$$\text{Lesser Outlier} = 56 - 1.5 * 26.5 = 56 - 39.75 = 16.25$$

$$\text{Greater Outlier} = 82.5 + 1.5 * 26.5 = 82.5 + 39.75 = 121.75$$

NIGHT:

$$IQR = 89 - 78 = 11$$

$$\text{Lesser Outlier} = 78 - 1.5 * 11 = 78 - 16.5 = 61.5$$

$$\text{Greater Outlier} = 89 + 1.5 * 11 = 89 + 16.5 = 105.5$$

IQR Summary Table

	Lesser Outlier	Minimum	Q_1	Median	Q_2	Maximum	Greater Outlier
Day	16.25	32	56	74.5	82.5	99	121.75
Night	61.5	25.5	78	81	89	98	105.5

Insights

- **Day:** No outliers detected — all values fall within the IQR range (16.25 to 121.75).
- **Night:** The minimum value (**25.5**) lies below the lower threshold (**61.5**), indicating a **lower outlier**. There are **no upper outliers** in the dataset.

INTERQUANTILE RANGE – PLACEMENT

[26]:

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108.0	67.303395	66.333163	66.370186	72.100558	62.278186	288655.405405
Median	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Mode	1	62.0	63.0	65.0	60.0	56.7	300000.0
Min	1.0	40.89	37.0	50.0	50.0	51.21	200000.0
Q1-25%	54.5	60.6	60.9	61.0	60.0	57.945	240000.0
Q2-50%	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Q3-75%	161.5	75.7	73.0	72.0	83.5	66.255	300000.0
99th	212.86	87.0	91.86	83.86	97.0	76.1142	NaN
Q4-100%	215.0	89.4	97.7	91.0	98.0	77.89	940000.0
IQR	107.0	15.1	12.1	11.0	23.5	8.31	60000.0
1.5rule	160.5	22.65	18.15	16.5	35.25	12.465	90000.0
Lesser	-106.0	37.95	42.75	44.5	24.75	45.48	150000.0
min	1.0	40.89	37.0	50.0	50.0	51.21	200000.0
max	215.0	89.4	97.7	91.0	98.0	77.89	940000.0
Greater	322.0	98.35	91.15	88.5	118.75	78.72	390000.0

Insights

- **Ssc_p**: No outliers detected — all values fall within the IQR range (37.95 to 98.35).
- **Hsc_p**: The minimum value (**37.0**) lies below the lower threshold (**42.75**), indicating a **lower outlier**. The maximum value (**97.7**) lies above the upper threshold (**91.15**), indicating an **upper outlier**.
- **Degree_p**: The maximum value (**91.0**) lies above the upper threshold (**88.5**), indicating an **upper outlier**. There are **no lower outliers** in the dataset.
- **Etest_p**: No outliers detected — all values fall within the IQR range (24.75 to 118.75).
- **Mba_p**: No outliers detected — all values fall within the IQR range (45.48 to 78.72).
- **Salary**: The maximum value (**940000.0**) lies above the upper threshold (**390000.0**), indicating an **upper outlier**. There are **no lower outliers** in the dataset.